

Merton's cumulative-advantage idea has a more general precursor, "preferential attachment," which, reversing the chronology (though not the logic), I will present next. Merton was interested in the social aspect of knowledge, not in the dynamics of social randomness, so his studies were derived separately from research on the dynamics of randomness in more mathematical sciences.

Lingua Franca

The theory of preferential attachment is ubiquitous in its applications: it can explain why city size is from Extremistan, why vocabulary is concentrated among a small number of words, or why bacteria populations can vary hugely in size.

The scientists J. C. Willis and G. U. Yule published a landmark paper in *Nature* in 1922 called "Some Statistics of Evolution and Geographical Distribution in Plants and Animals, and Their Significance." Willis and Yule noted the presence in biology of the so-called power laws, attractive versions of the scalable randomness that I discussed in [Chapter 3](#). These power laws (on which more technical information in the following chapters) had been noticed earlier by Vilfredo Pareto, who found that they applied to the distribution of income. Later, Yule presented a simple model showing how power laws can be generated. His point was as follows: Let's say species split in two at some constant rate, so that new species arise. The richer in species a genus is, the richer it will tend to get, with the same logic as the Mathew effect. Note the following caveat: in Yule's model the species never die out.

During the 1940s, a Harvard linguist, George Zipf, examined the properties of language and came up with an empirical regularity now known as Zipf's law, which, of course, is not a law (and if it were, it would not be Zipf's). It is just another way to think about the process of inequality. The mechanisms he described were as follows: the more you use a word, the less effortful you will find it to use that word again, so you borrow words from your private dictionary in proportion to their past use. This explains why out of the sixty thousand main words in English, only a few hundred constitute the bulk of what is used in writings, and even fewer appear regularly in conversation. Likewise, the more people aggregate in a particular city, the more likely a stranger will be to pick that city as his destination. The big get bigger and the small stay small, or get relatively smaller.

A great illustration of preferential attachment can be seen in the mushrooming use of English as a lingua franca—though not for its intrinsic qualities, but because people need to use one single language, or stick to one as much as possible, when they are having a conversation. So whatever